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MILITARY SPECIFICATION

AIR CLEANER, ENGINE: HEAVY-DUTY, OIL-BATH TYPE
(FOR INTERNAL-COMBUSTION ENGINES)

1. SCOPE

1.1 This specification covers heavy-duty air cleaners of the oil-bath type for use with internal-combustion engines.

1.2 Classification. Air cleaners shall be furnished in the following types (see 6.2).

Type I	-	For automotive applications.
Type II	-	For industrial applications.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-F-506	-	Flux, Soldering; Paste and Liquid.
P-D-680	-	Dry Cleaning Solvent.
QQ-S-571	-	Solder: Lead Alloy, Tin Lead Alloy, and Tin Alloy; Flux Cored Ribbon and Wire, and Solid Form.

Military

MIL-T-704	-	Treatment and Painting of Material.
MIL-L-2104	-	Lubricating Oil, Internal-Combustion Engine, Heavy Duty.

FSC-2940

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MIL-L-10295 - Lubricating Oil, Internal-Combustion Engine, Sub-Zero.
MIL-D-13570 - Dust, Testing by Exposure to.

STANDARDS
Federal
FED. STD. NO. 595 - Colors.

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MS 35875 - Air Cleaner, Intake; Oil Bath Type, for Industrial Engines.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement and functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article sample. A first article sample of the air cleaner, to be furnished under this specification, shall be produced prior to the manufacture of this item in production quantity. The first article sample of the air cleaner shall be submitted to the Government for first article inspection to determine conformance to the quality assurance provisions of this specification (see 4.2). The first article sample of the air cleaner submitted by a contractor shall be fully representative of the material to be supplied from production facilities and tooling.

3.2 Materials. Materials shall be as specified herein and in the applicable drawings or standards. Materials or equipment not designated on the drawings or standards and not specifically covered by this or other specifications shall be uniform in quality and free from imperfections or defects which affect their serviceability.

3.2.1 Solder. Solder, when used, shall conform to specification QQ-S-571.

3.2.1.1 Soldering flux. Soldering flux, when used, shall conform to specification O-F-506.

3.2.2 Sheet metal parts. Sheet metal parts shall be free of pipes, blisters, tears, and excessive metal thinning at drawn sections.

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3.2.3 Filtering element. Filtering elements shall be free of loose materials.

3.3 Design and construction.

3.3.1 Interchangeability. All units and their replaceable components shall be physically and functionally interchangeable with all other such units of the same military part number previously furnished for this application.

3.3.2 Servicing. Types I and II air cleaners shall be so designed and constructed that servicing (including the removal of the oil cup or removable filter elements if used) cleaning, replenishing of the oil supply, and reassembly may be accomplished without the use of special tools and without removing or disturbing the clean air chamber or its connection to the engine. The quantity of parts required to be removed for servicing shall be held to a minimum and shall be so designed as to prevent improper reassembly. Design shall permit replenishment of oil without removal of parts other than the oil cup or oil cup and precleaner when precleaner is part of oil cup assembly.

3.3.3 Filtering element. The filter element of the air cleaner shall incorporate a fixed element and a removable element or the complete element shall be removable. The removable element shall be the upstream portion of the filter element and shall be self-contained subassembly held securely in place when the air cleaner is completely assembled. A removable element shall be optional in type II air cleaners of up to 135 cubic feet per minute (cfm) rated capacity and shall be required for type II air cleaners above 135 cfm capacity.

3.3.4 Oil level marking. The proper oil level shall be clearly indicated by a permanent marking. The marking shall be clearly visible in ordinary light at the time the oil is being replenished. The marking shall be so located as to minimize the possibility of filling to an improper level.

3.3.5 Tubular joints. Unless otherwise specified (see 6.2), only the air cleaner outlet tubular joint shall incorporate an externally beaded outer end. Tubular joints shall be capable of withstanding external pressures caused by hose clamps or other fastening devices of similar purpose.

3.3.6 Flanged joints. Flanged joints shall have smooth faces suitable for use with gaskets, and when assembled, shall provide airtight joints under all conditions of service. There shall be no holes in external flanges opening into the clean-air passage which will permit leakage of air.

3.3.7 Rated air flow. Design shall be such that rated air flow capacity in cubic feet per minute of air shall be as specified on the applicable drawings or by the procuring activity (see 6.2). Unless otherwise specified, the rated air flow for types I and II air cleaners shall be the capacity rating selected from the column headed "RATED AIR CLEANER AIR FLOW (CFM)" in standard MS35875 for the applicable MS part number (see 6.2).

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3.4 Performance. The air cleaner shall have the performance characteristics specified herein when serviced and filled with the amount of oil specified (see 3.3.4) for the applicable operating temperature ranges specified in table I.

Table I. Oil

Specification	SAE grade	Ambient air temperature range, degrees F (approx.)
MIL-L-2104	30	Above +32°F
MIL-L-2104	10	+40° to -10°F
MIL-L-10295		0° to -65°F

3.4.1 Air leakage. When tested, as specified in 4.5.3.2, there shall be no evidence of air leakage.

3.4.2 Backfire. All air cleaners used on gasoline engines shall not lose oil in excess of 0.1 percent of the rated oil capacity and shall sustain no structural damage or permanent deformation when tested as specified in 4.5.3.3.

3.4.3 Air flow restriction. Restriction of a serviced type I air cleaner, at rated air-flow and operated without oil being pulled over, shall not exceed 12 inches of water if intended for use on a carbureted engine, and 15 inches of water if intended for use on a diesel engine. For type II air cleaners, this restriction shall not exceed 12 inches of water. This restriction shall be interpreted as the static-pressure drop across the cleaner. Testing for air flow restriction shall conform to 4.5.3.4.

3.4.4 Oil pullover.

3.4.4.1 High oil level. When the oil level is 20 percent by weight above normal level, the type I air cleaner shall have sufficient air capacity to operate for not less than 15 minutes at 110 percent of rated air flow for a carbureted engine, or 150 percent of rated air flow for a diesel engine, without oil being pulled over, when tested as specified in 4.5.3.5.1. For type II air cleaners, the above requirements shall apply at 150 percent of rated air flow.

3.4.4.2 Angle operation. The air cleaner shall not pull oil over during 15 minutes of operation at 110 percent of rated air flow (for type I carbureted engine use) or at 150 percent of rated air flow (for type I diesel engine use or all type II air cleaners) when the cleaner is tipped at an angle of 30 degrees from vertical toward the air outlet as specified in 4.5.3.5.2. Oil in cleaner shall be at normal level before test is started.

3.4.5 Efficiency.

3.4.5.1 Type I. The type I air cleaner, at rated air-flow, shall have an efficiency of not less than 98 percent and not less than 96% at 25% of rated air flow when tested as specified in 4.5.3.6.1. The efficiency at varying air flow shall not be less than 97 percent when tested as specified in 4.5.3.6.2.

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3.4.5.2 Type II. The type II air cleaner, at rated air-flow, shall have an efficiency of not less than 98 percent and not less than 96 percent at 25 percent of rated air flow when tested as specified in 4.5.3.6.1.

3.4.6 Dust capacity.

3.4.6.1 Type I. The type I air cleaner at normal oil level, operated with varying air-flow without exceeding a restriction of 20 inches of water at rated air-flow, shall have a total capacity of coarse dust of not less than 18 grams for each cfm of rated air-flow for a carbureted engine, and 21 grams for each cfm of rated air-flow for a diesel engine, when tested as specified in 4.5.3.7.

3.4.6.2 Type II. The type II air cleaner shall meet the requirements of 3.4.6.1 with a total capacity of coarse dust of not less than 18 grams for each cfm of rated air flow (see MS35875). In addition, the type II air cleaner, with weight W (see 4.5.3.1.2) of oil in the cup, operated with a constant air flow of 75 percent of rated air flow, and a coarse-dust concentration of 0.050 grams for each cfm actual air flow (75 percent of rated air-flow) shall have a restriction of not more than 20 inches of water when tested as specified in 4.5.3.8.

3.4.7 Restriction after dust capacity test. After servicing the oil cup and removable element following the dust-capacity test, the cleaner restriction at rated air-flow shall not exceed its original restriction plus 2 inches of water when tested as specified in 4.5.3.7.1 or 4.5.3.8.1.

3.4.8 Oil loss (type I). The type I air cleaner shall show no evidence of oil loss under all conditions of operation, provided that the cleaner has been properly serviced. The decrease in weight of oil shall not exceed 0.5 percent of the weight of oil in the oil cup at beginning of the test, after the air cleaner has been operated 3 hours at rated air-flow when tested as specified in 4.5.3.9.

3.4.9 Vibration resistance. When tested, as specified in 4.5.3.10, the air cleaner shall show no evidence of structural damage or permanent deformation.

3.5 Finish.

3.5.1 Tactical. Unless otherwise specified (see 6.2), treatment and painting shall be in accordance with type A of specification MIL-T-704.

3.6 Color. The exterior color shall be as specified by the procuring activity (see 6.2) and shall be in accordance with standard FED. STD. NO. 595.

3.7 Marking. Marking shall be in accordance with standard MIL-STD-130 and shall include the rated capacity (see 3.3.7).

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3.8 Workmanship. Workmanship shall be such that the air cleaners shall be free of burs, sharp edges, surfaces out of alignment or out of contour (visually apparent); paint sags, runs, blisters or incomplete coverage; loose metal particles or other foreign materials.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 First article inspection. A first article sample consisting of one air cleaner of each type and capacity (rated four, cfm) shall be submitted for first article inspection. The samples shall be representative of the products proposed to be furnished under the contract. First article inspection shall be conducted by the Government at a place designated by the procuring activity and shall consist of examination for the defects specified and 4.3.2.2 and testing as specified in table II. No changes shall be made in the product after the performance of first article inspection except as specifically approved by the procuring activity.

4.2.1 Failure. Failure of the first article sample to pass any of the specified inspection may be cause for the Government to refuse to perform additional inspection until the faults revealed have been corrected.

Table II. First article tests

Paragraph Number	Test
4.5.3.2	Air leakage
4.5.3.3	Backfire
4.5.3.4	Air flow restriction
4.5.3.5.1	Oil pullover at high oil level
4.5.3.5.2	Oil pullover at angle operation
4.5.3.6.1	Efficiency at rated air flow
4.5.3.6.2	Efficiency at varying air flow (type I only)
4.5.3.7	Dust capacity with varying air flow
4.5.3.7.1	Restriction after variable flow dust capacity test
4.5.3.8	Dust capacity with constant air flow (type II only)
4.5.3.8.1	Restriction after constant flow dust capacity test (type II)
4.5.3.9	Oil loss (type I)
4.5.3.10	Vibration

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4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 Lot formation. Unless otherwise specified, a lot shall consist of all air cleaners of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.3.1.2 Quality conformance examination. Samples for quality conformance examination shall be selected in accordance with Standard MIL-STD-105.

4.3.1.3 Quality conformance testing. Sampling for quality conformance testing shall be selected in accordance with inspection level S-3 of the appendix to Standard MIL-STD-105.

4.3.2 Examination inspection for air cleaners.

4.3.2.1 Acceptable quality level. Each sample selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL) on basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.3.2.2 Classification of defects. For examination purposes, defects shall be classified as follows:

Categories		
	Major defects (1.0 AQL)	Examination method
101	Defective sheet metal (see 3.2.2)	Visual and gage
102	Filter element with loose material (see 3.2.3)	Visual
103	Incorrect dimensions affecting interchangeability (see 3.3.1)	Scale or gage
104	Incorrect servicing provisions (see 3.3.2)	Visual
105	Filter element of improper design (see 3.3.3)	Visual
106	Missing or incorrect oil-level marking (see 3.3.4)	Visual
107	Bent or otherwise defective tubular and flanged joints (see 3.3.5 and 3.3.6)	Visual
	Minor defects (2.5 AQL)	Examination method
201	Incorrect dimensions not affecting interchangeability (see 3.3.1)	Scale or gage
202	Loose filter element (see 3.3.3)	Visual
203	Improper finish (see 3.5)	Visual
204	Improper color (see 3.6)	Visual
205	Missing or incorrect marking (see 3.7)	Visual
206	Poor workmanship (see 3.8)	Visual

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4.3.3 Testing of air cleaners.

4.3.3.1 Quality conformance tests. Samples selected in accordance with 4.3.1.3 shall be subjected to the tests specified in table III, in the order listed, using an AQL of 6.5 on the basis of percent defective.

Table III. Order of quality conformance tests

Test	Paragraph Number
Backfire	4.5.3.3
Air leakage	4.5.3.2
Air flow restriction	4.5.3.4
Oil pullover at high oil level	4.5.3.5.1
Oil pullover at angle installation	4.5.3.5.2
Oil loss (type I only)	4.5.3.9
Vibration	4.5.3.10
Air leakage	4.5.3.2

4.3.4 Inspection of preparation for delivery. The cleaning, packaging, packing and marking of the air cleaners shall be inspected to determine compliance with section 5 and the specifications referenced therein.

4.4 Control tests.

4.4.1 Sampling for control tests. Control tests samples shall be selected at the rate of 3 of each 500 air cleaners produced except that not less than 3 nor more than 6 air cleaners shall be selected in any 30-day period.

4.4.2 Tests. Air cleaners selected in accordance with 4.4.1 shall be examined for the defects specified in 4.3.2.2 and subsequently tested as specified in table IV.

Table IV. Control tests

Sample Number	Test	Paragraph Number
1	Efficiency at rated air flow	4.5.3.6.1
	Efficiency at varying air flow (type I)	4.5.3.6.2
2	Dust capacity with variable air flow	4.5.3.7
	Restriction resulting from variable air flow dust capacity test	4.5.3.7.1
3	Dust capacity with constant air flow (type II only)	4.5.3.8
	Restriction after constant air flow dust capacity test (type II only)	4.5.3.8.1

4.4.3 Failure. Failure of a control test sample to pass any specified examination or test may be cause for the Government to refust to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the test have been corrected.

4.5 Tests.

4.5.1 Test conditions.

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4.5.1.1 Temperature. Ambient temperature during tests shall be 80° ± 10°F.

4.5.1.2 Relative humidity. Relative humidity during tests shall be 35 ± 5 percent.

4.5.1.3 Air flow. Specified values of air flow are for standard atmospheric conditions of 29.92 inches of mercury barometric pressure and 80°F ambient temperature. Actual air flow values shall be corrected to standard conditions.

4.5.1.4 Test dust.

4.5.1.4.1 Fine. Fine test dust shall be as specified in specification MIL-D-13570.

4.5.1.4.2 Coarse. The chemical content of coarse test dust shall be as specified in specification MIL-D-13570. Particle size distribution by weight shall be as follows:

Size in microns	Percent of total weight
0-5	12 ± 2
5-10	12 ± 2
10-20	14 ± 3
20-40	23 ± 3
40-80	30 ± 3
80-200	9 ± 3

4.5.1.4.3 Dust feeding rate. When the feeding rate of dust is required as part of any test, the dust shall be fed at a uniform rate of 0.025 gram per cubic foot of air unless otherwise specified.

4.5.1.5 Test oil. Oil used for tests shall conform to grade 30 of specification MIL-L-2104.

4.5.2 Test equipment.

4.5.2.1 Piezometer, adapter, and manometer. Piezometers for tests specified herein shall be selected from figure 1. Dimension D shall be equal to or the next size smaller than the air cleaner outlet inside diameter. Adapters for the applicable piezometers shall also be selected from figure 1. Water manometers shall be installed as shown in figure 2 as applicable.

4.5.2.2 Flowmeter. A flowmeter of adequate capacity shall be connected to the outlet of the adapter.

4.5.2.3 Absolute cleaner. The absolute cleaner shall consist of two layers of 10-ounce canton flannel installed in a circular frame in such a manner as to filter all air passing through the outlet of the air cleaner (see figure 3).

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4.5.2.3.1 Preparation for use. The canton flannel shall be hung in an air oven and dried for 6 hours at a temperature of 230° ± 10° and then weighed to the nearest 0.01 gram. It shall then be placed in a dust and moisture proof container until ready for use.

4.5.3 Test procedures.

4.5.3.1 Run-in procedure. Preparatory to the start of any test, the air cleaner shall be run-in as specified herein.

4.5.3.1.1 Washing and drying of air cleaner. The air cleaner shall be disassembled and the oil cup, body, and removable element shall be drained and subsequently cleaned with a solvent conforming to specification P-D-680. After cleaning, the cup and body shall be wiped dry and the removable element dried by forcing air through it to remove solvent, or by the use of a drying oven.

4.5.3.1.2 Preparation. The oil cup of the cleaner, without oil, shall be weighed and the weight recorded. The oil cup shall then be filled to normal level, by sight, weighed and the weight recorded. The weight (W) of the oil in the cup shall then be calculated to the nearest 0.1 gram and recorded. Subsequent filling of the oil cup shall be made using the weight of the oil thus determined.

4.5.3.1.3 Pretest operation. The oil cup shall be filled with W ± 0.1 grams of test oil and assembled to the air cleaner. The air cleaner shall be installed on a test stand as shown in figure 2, as applicable, and operated at rated flow for 15 minutes. Restriction (differential pressure across the air cleaner)shall be read and recorded at 5-minute intervals.

4.5.3.1.4 Oil replenishment. After pretest operation, the oil cup shall be immediately removed and its oil replenished to weight (W) (see 4.5.3.1.2). With the air cleaner in normal position and the oil cup removed, the elements shall be allowed to drain for 15 minutes. The oil cup shall then be replaced, the elapsed time being not more than 17 minutes after completion of the pretest operation (see 4.5.3.1.3).

4.5.3.2 Air leakage. The air cleaner body and oil cup sealing surfaces shall be cleaned prior to assembly. The air cleaner shall be assembled and installed, with one port sealed, on the test stand as shown in figure 2. The air cleaner shall be evacuated to 80 ± 1.5 inches of water vacuum and the shut-off valve closed. A manometer reading shall be taken after closing the valve and again 5 ± 1/2 minutes later to determine conformance to 3.4.1.

4.5.3.3 Backfire. The air cleaner shall be operated as specified in 4.5.3.1 and subsequently connected to the apparatus shown in figure 4. With air chamber pressure at 5 psi, and both valves closed, the stopcock shall be suddenly opened. This procedure shall be repeated five times at each of the following pressures: 5, 10, 15, 20, 25, 30, and 40 psi. The total loss of oil shall be recorded and the air cleaner examined for structural damage to determine conformance to 3.4.2.

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4.5.3.4 Air flow restriction. The air cleaner shall be operated, as specified in 4.5.3.1.3, and subsequently installed on the test stand shown in figure 3. Air flow shall be set at 25 percent of rated flow and the restriction recorded after 5 minutes of operation. This procedure shall be repeated at 50, 75, 100, and 150 percent of rated flow. The air cleaner shall be observed throughout the test for evidence of oil pullover to determine conformance to 3.4.3.

4.5.3.5 Oil pullover.

4.5.3.5.1 High oil level. The air cleaner shall be run, as specified in 4.5.3.1.3, and the oil level increased to 20 percent above normal by weight. The air cleaner shall subsequently be connected as shown in figure 1 with a piezometer attached to the adapter and the air flow started and adjusted to 110 percent of rated air flow (150 percent for diesel engines) in increments of 10 percent of rated air flow (expressed as the nearest multiple of 10 cfm). Adjustment shall be accomplished once every 1 1/2 minutes and the restriction recorded each time prior to an adjustment to determine conformance to 3.4.5. The air cleaner shall also be operated at 110 or 150 percent of maximum air flow for 15 minutes and the restriction recorded once every 1 1/2 minutes.

4.5.3.5.2 Angle. Prior to testing, the cleaner shall be operated as specified in 4.5.3.1.3. The air cleaner shall subsequently be connected as specified in 4.5.3.5.1, except oil level shall be at normal level, tilted 30 degrees on its vertical axis in such a manner that the outlet is on the lower side, and operated as specified in 4.5.3.5.1 to determine conformance to 3.4.4.2.

4.5.3.6 Efficiency.

4.5.3.6.1 Efficiency at rated air flow. The air cleaner shall be run in as specified in 4.5.3.1.3. An absolute cleaner shall be installed as specified in 4.5.2.3 and the air flow started and adjusted to rated flow. Fine test dust (see 4.5.1.4.1)shall be fed (see figure 5)into the air cleaner inlet at the rate of 0.025 grams of dust per cubic foot of air until 110 grams of dust have been fed or until 30 minutes of feeding shall have elapsed. The absolute cleaner shall be removed, taking care to retain all entrapped dust, and the efficiency obtained by means of the following formula to determine conformance to 3.4.5:

$$\text{Efficiency, percent} = \frac{W_2 - W_3}{W_1} \times 100$$

Where W_1 = Weight of dust fed into air cleaner inlet
W_2 = Weight of absolute cleaner and entrapped dust
W_3 = Original weight of absolute cleaner

NOTE: All measurements to be accurate to nearest 0.01 gram

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4.5.3.6.2 Efficiency at varying air flow (type I only). The air cleaner shall be run in as specified in 4.5.3.1. An absolute cleaner shall be installed as specified in 4.5.2.3 and air flow started and adjusted to rated air flow. The air flow shall be varied as shown in figure 6. Fine dust (see 4.5.1.4.1) shall be fed (see 4.5.1.4.3) into the air cleaner inlet at 60 percent of rated air flow. Not less than 110 grams of dust shall be fed. Should the time required to feed 110 grams of dust be less than 30 minutes, dust shall be fed for 30 minutes at the above concentration, and the amount of dust recorded. The absolute cleaner shall be removed, taking care to retain all entrapped dust, and the efficiency obtained as in 4.5.3.6.1 to determine conformance to 3.4.5.

4.5.3.7 Dust capacity with variable air flow (types I and II). Prior to start of test, the air cleaner shall be run in as specified in 4.5.3.1.3, the absolute cleaner installed as specified in 4.5.2.3, and the dust feeder filled with a weighed amount of coarse test dust (see 4.5.1.4.2). Air flow shall be started 1 minute prior to start of test and adjusted to rated air flow. Restriction across the air cleaner shall be recorded. At start of test, the variable air cycle as shown in figure 6 shall be started and the feeding of test dust into the air cleaner inlet started at a constant rate of (F) grams per hour where:

F = 0.9 x rated air flow in cubic feet per minute (Cfm).

The air flow cycle shall be repeated until three successive air flow restrictions exceed 20 inches of water or a restriction of 24 or more inches of water is reached. At each test stop (see figure 7), dust feeder shall be weighed before and after filling with test dust. At test stop prior to each efficiency test (see figure 7), an absolute air cleaner, as specified in 4.5.2.3, shall be installed. At test stop after each efficiency test, the absolute cleaner shall be removed, dried, weighed, and the efficiency determined as in 4.5.3.6. All test dust shall be returned to the dust feeder. Testing shall continue until the third air flow restriction value above 20 inches of water or the first restriction value of 24 inches of water or above, whichever occurs first, is attained. The total weight of dust shall subsequently be calculated to determine conformance to 3.4.6.1 and 3.4.6.2.

4.5.3.7.1 Restriction after variable flow dust capacity test (types I and II). The oil cup and removable element shall be removed and cleaned after completion of 4.5.3.7. The oil cup shall be refilled to weight W (see 4.5.3.1.2) and reassembled to the air cleaner. The air flow shall be started, adjusted to rated flow, and the restriction across the air cleaner recorded. The air flow shall be varied as in 4.5.3.7 and the restriction recorded at the start of each cycle and at the end of 30 minutes of operation (end of test) to determine conformance to 3.4.7.

4.5.3.8 Dust capacity with constant air flow (type II only). Prior to the start of test, the air cleaner shall be run in, the absolute cleaner installed and the dust feeder filled as specified in 4.5.3.7. Restriction at rated air flow shall be recorded. Air flow shall be reduced to 75 percent of rated flow and dust shall be fed into the air cleaner inlet at a constant rate of G grams per hour where:

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G = 2.25 x rated air flow in cubic feet per minute (Cfm).

Restriction shall be recorded at 10-minute intervals. This procedure shall be continued until (a) 6 hours of testing have elapsed or (b) three successive readings of restriction have been in excess of 20 inches of water, whichever should occur first, to determine conformance to 3.4.6.2.

4.5.3.8.1 Restriction after constant flow dust capacity test. The oil cup and removable element shall be removed and cleaned after completion of 4.5.3.8. Oil cup shall be refilled to weight W (see 4.5.3.1.2) and reassembled to the air cleaner. Air flow at rated flow shall be maintained for 30 minutes and the restriction recorded at 0, 10, 20, and 30 minutes of elapsed time, to determine conformance to 3.4.7.

4.5.3.9 Oil loss (type I). The type I air cleaner shall be run in as specified (see 4.5.3.1). The assembled air cleaner shall be wieghed to the nearest 0.1 gram and the weight recorded. The air cleaner shall then be connected to the flowmeter inlet as shown on figure 2. Air flow shall be started and adjusted to rated air flow and the air cleaner operated for 3 hours. At 1/2 hour intervals, restriction, air flow, relative humidity, barometric pressure, and temperature shall be recorded. The transparent piezometer shall be observed for traces of oil. At the end of the 3 hour period, the air flow shall be stopped and the air cleaner disconnected. The air cleaner shall be reweighed and weight recorded. Percentage of oil loss shall be determined from the following equation to determine conformance to 3.4.8:

$$\frac{X-Y}{Z} \times 100 = \text{percent of oil loss}$$

Where X = total weight of air cleaner at start (with oil)
Y = total weight of air cleaner after test (with remaining oil)
Z = weight of oil in air cleaner at start of test"

4.5.3.10 Vibration. To determine conformance to 3.4.9, the air cleaner shall be mounted as in intended operation and subjected to method 201A vibration test of standard MIL-STD-202, except that frequency shall be varied between 10 and 40 cycles per second. Vibration shall be applied horizontally in the direction of the longitudinal centerline of the air outlet. The air cleaner shall be vibrated for 24 hours and any structural or assembly breakdown during this time interval shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with the applicable packaging standard or packaging data sheet specified by the procuring activity (see 6.2).

6. NOTES

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6.1 Intended use. The air cleaners covered by this specification are intended for use with internal combustion engines for automotive and industrial applications.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type (see 1.2).
(c) Applicable MS part number (see 3.3.7).
(d) Rated air flow if not specified on drawing (see 3.3.7).
(e) Protective finish if other than that specified (see 3.5.1).
(f) Color required (see 3.6).
(g) Selection of applicable packaging data sheet (see 5.1).
(h) Where first article inspection will be conducted.
(i) Beaded outer end required for other than cleaner outlet joint.

6.3 First article inspection. Unless otherwise specified (see 6.2), first article inspection will be conducted by U. S. Army Tank Automotive Command, ATTN: AMSTA-RKA, Warren, Michigan 48090.

6.4 Certain provisions of this specification are the subject of international standardization agreement (STANAG 4049). When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodian:
Army - AT

Preparing activity:
Army - AT

Project No. 2940-A085

User:
CS

International interest (see sect. 6)

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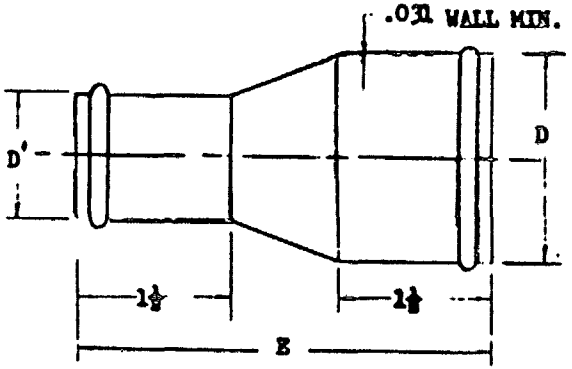


FIGURE 1A
ADAPTER

D	E	D'	D	E	D'	D	E	D'
2	3-7/32	1-3/4	3	3-7/32	2-3/4	4	3-7/32	3-3/4
2	2-7/16	1-1/2	3	3-7/16	2-1/2	4	3-7/16	3-1/2
2	3-5/8	1-1/4	3	3-5/8	2-1/4	4	3-5/8	3-1/4
2	3-7/8	1	3	3-7/8	2	4	3-7/8	3
2	4-1/8	3/4	3	4-1/8	1-3/4			
			3	4-5/16	1-1/2			
			3	4-17/32	1-1/4			

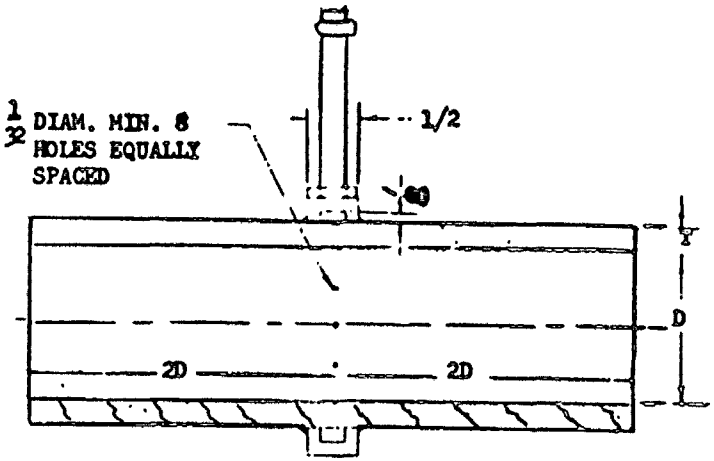


FIGURE 1B
PIEZOMETER

Nominal size	Transparent Plastic D(I.D.)	T	Steel D(O.D.)	T
1-1/2	1-1/2	1/4	1-1/2	.032
1-3/4	1-3/4	1/4	1-3/4	.035
2	2	1/4	2	.035
2-1/4	2-1/4	1/4	2-1/4	.035
2-5/16			2-5/16	.035
2-1/2	2-1/2	1/4	2-1/2	.035
2-5/8			2-5/8	.049
2-3/4	2-3/4	1/4	2-3/4	.049
3	3	1/4	3	.049
3-1/4			3-1/4	.065
3-1/2	3-1/2	1/4	3-1/2	.083
3-3/4			3-3/4	.083
4	4	1/4	4	.083
4-1/4			4-1/4	.083
4-1/2			4-1/2	.083

FIGURE 1 ADAPTER AND PIEZOMETER

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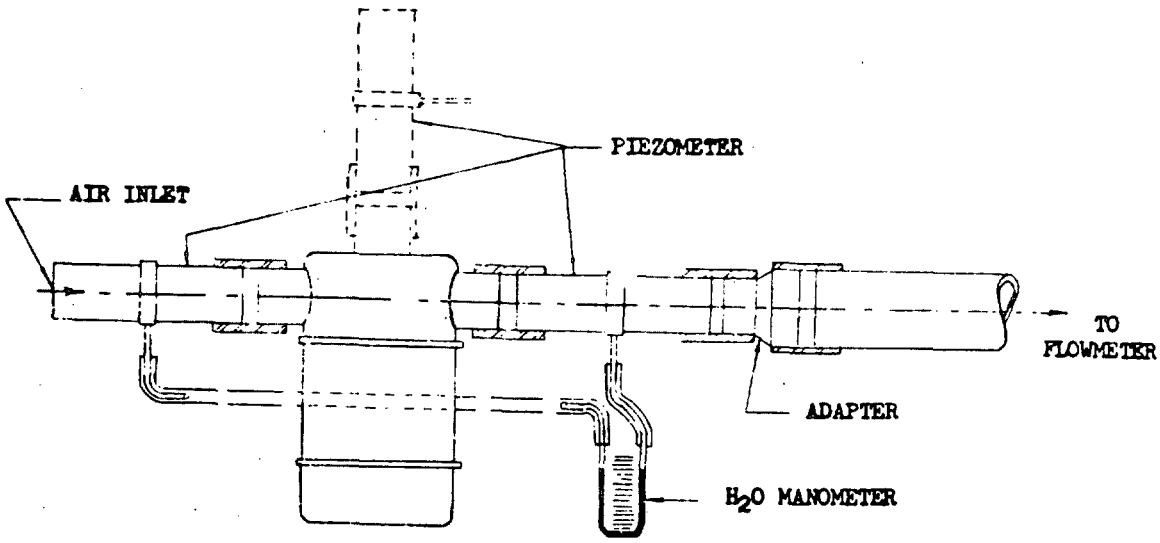


FIGURE 2A

TUBULAR INLET CLEANERS

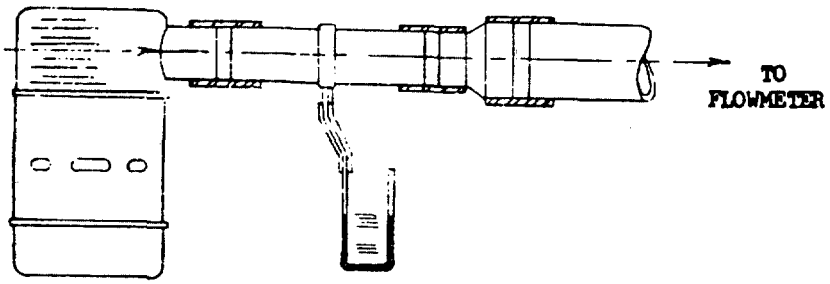


FIGURE 2B

CIRCUMFERENTIAL OR

LOUVERED INLET CLEANERS

FIGURE 2
TEST SET UP.

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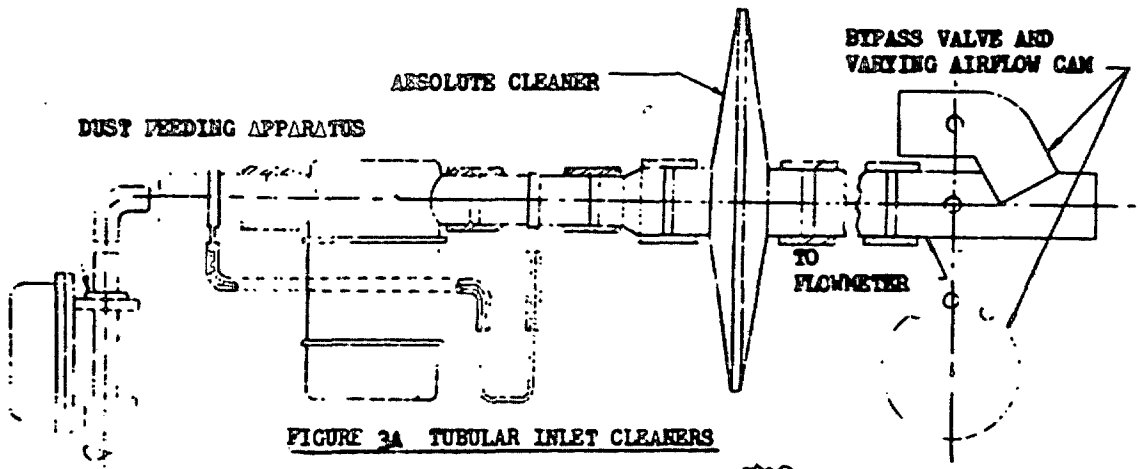


FIGURE 3A TUBULAR INLET CLEANERS

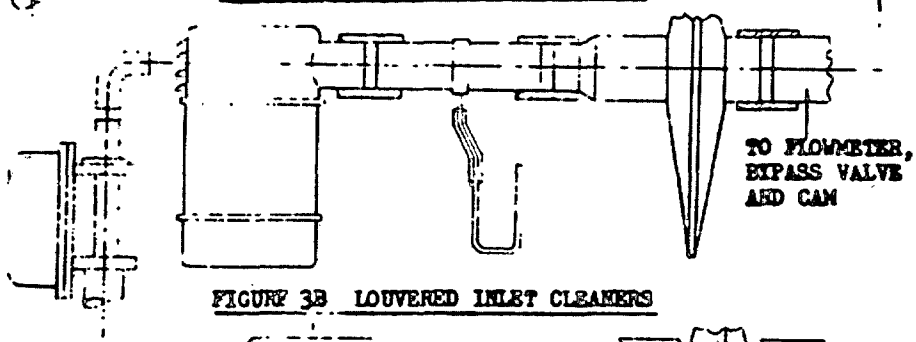


FIGURE 3B LOUVERED INLET CLEANERS

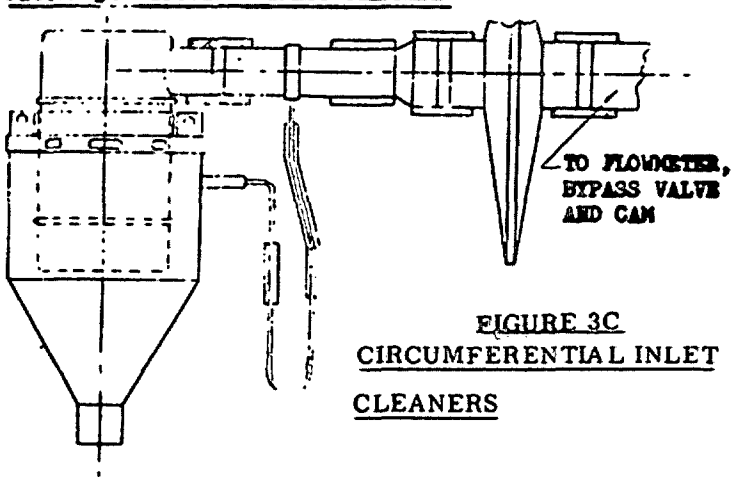


FIGURE 3C
CIRCUMFERENTIAL INLET
CLEANERS

FIGURE 3 TEST SET-UP

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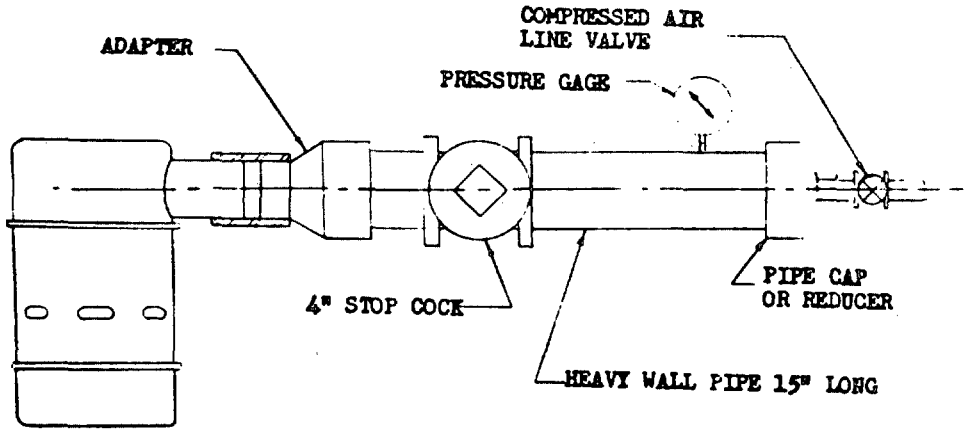


FIGURE 4

TEST SET-UP

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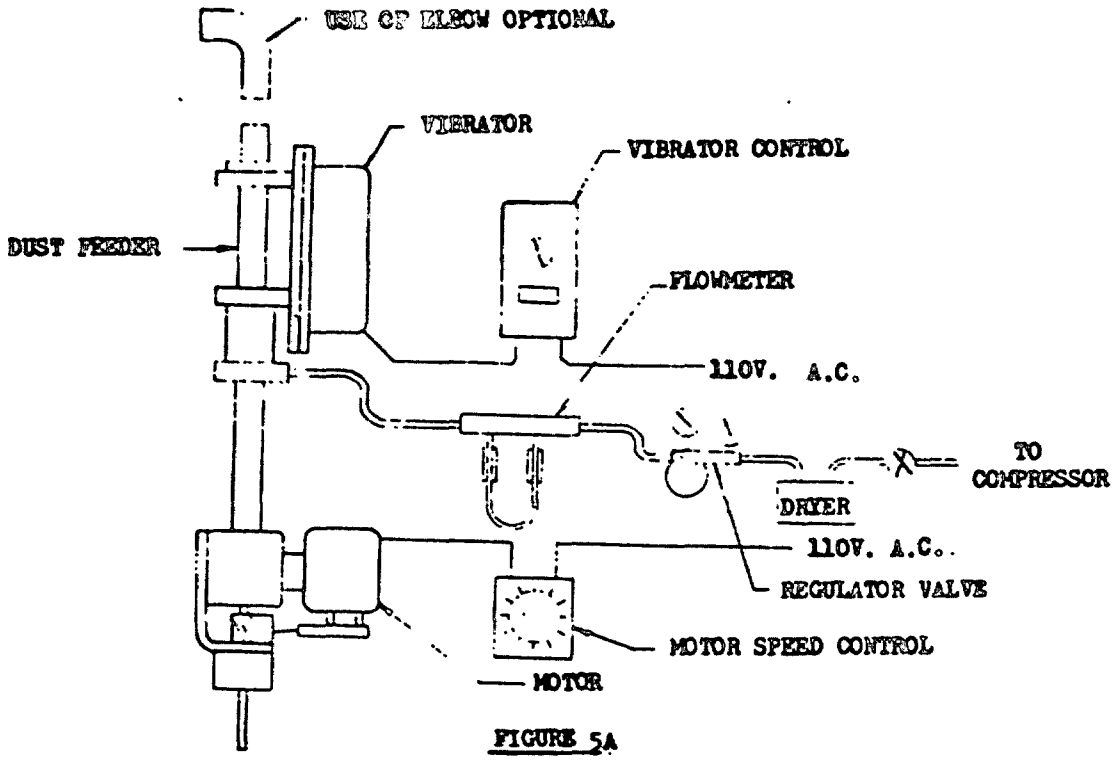


FIGURE 5A

DUST FEEDER

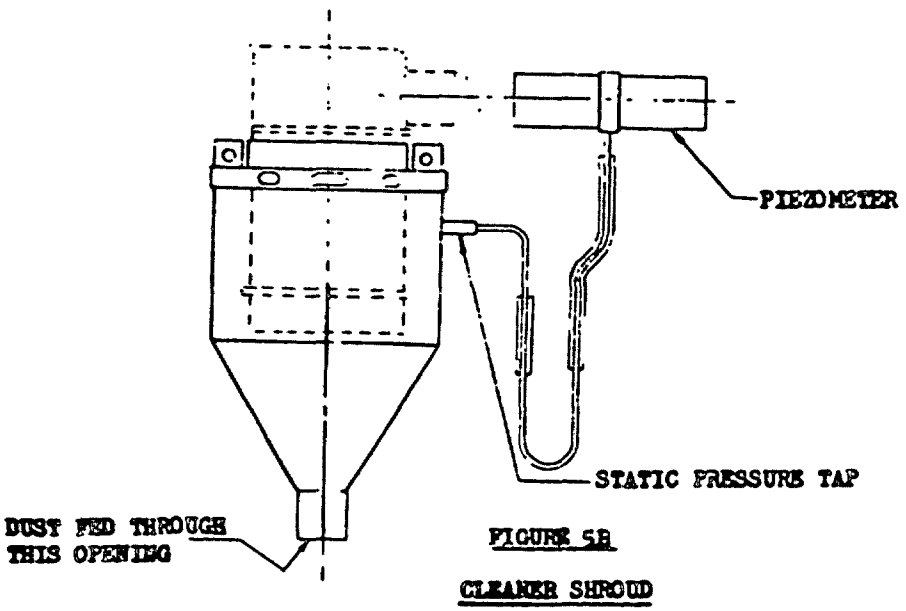


FIGURE 5B

CLEANER SHROUD

FIGURE 5 DUST FEEDING SET-UP

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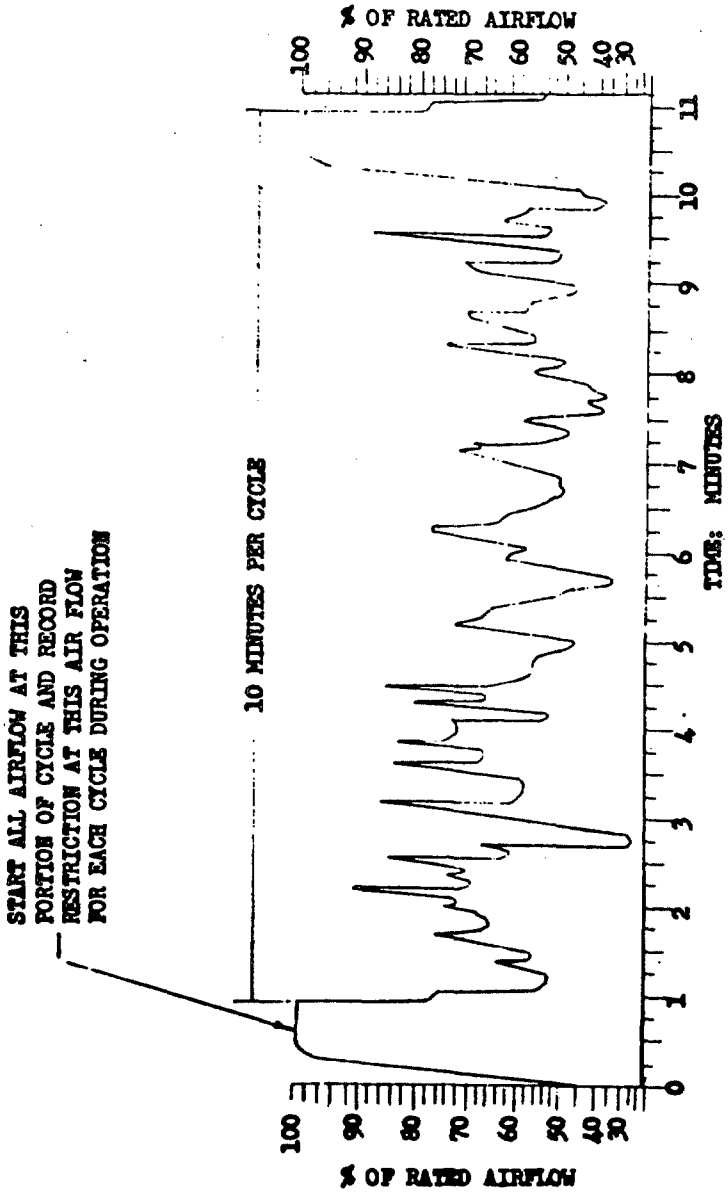


FIGURE 6
CYCLE OF AIRFLOW VARIATION

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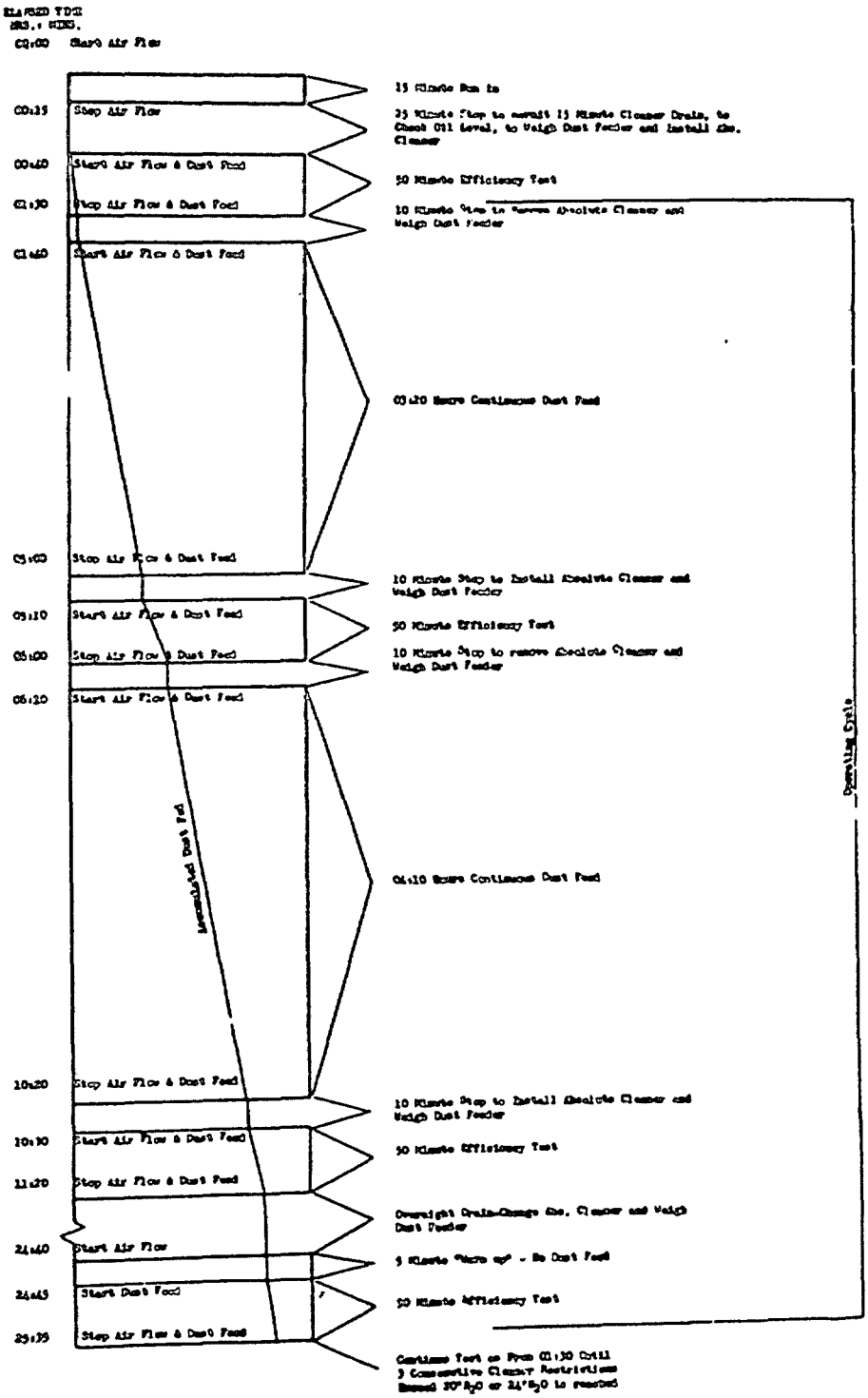


FIGURE 7
OPERATING CYCLE
OIL BATH AIR CLEANER DUST CAPACITY TEST

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